

Examiner reports

Q1.

The responses to this question were very mixed, varying from the very poor to some excellent answers. Most, but not all, candidates did attempt to separate and integrate in part (a)(i) with some making an error in their integrals, and rather more omitting a constant. Such candidates could make no further progress in part (a)(ii) without a constant to find. Some candidates thought to bring e^{-1} into their answer in part (a)(i), instead of the expected conventional constant, and got themselves confused in part (a)(ii) as they could not convincingly obtain the given answer. Some candidates made the anticipated error at

the exponentiation stage of $\ln y = -\cos t + c \Rightarrow y = e^{-\cos t} + e^c$. Many candidates could not justify the 50, other than saying it was the initial value.

For part (b) many candidates realised they had been given the required expression in part (a) and substituted and evaluated correctly, although some ignored the request to answer to the nearest centimetre, any greater accuracy being rather meaningless in this modelling problem. Some used degrees instead of radians. Other candidates substituted into the differential equation. In part (b)(ii), although most candidates knew in principle what to do, they could not find the second derivative correctly. Many treated y as a constant or as if it were t , rather than using implicit differentiation.

Some candidates did attempt to differentiate the given answer in part (a)(ii) twice, but usually made an error in their use of the chain rule. Some candidates did get the differentiation correct, but failed to complete their argument by demonstrating that the first derivative was zero at $t = \pi$, so confirming this point as a turning point.

Q2.

Part (a) Almost all candidates were able to find the first and second derivatives correctly, although there was an occasional arithmetic slip; some could not cope with the fraction term, others doubled 26 incorrectly.

Part (b) Those who substituted $t = 2$ into $\frac{dy}{dt}$ did not always explain that $\frac{dy}{dt} = 0$ is the condition for a stationary point. Many used the second derivative test and concluded that the point was a maximum. Some **assumed** that a stationary point occurred when $t = 2$ and went straight to the test for maximum or minimum and only scored half of the marks.

A few tested $\frac{dy}{dt}$ on either side of $t = 2$ correctly, but those who only considered the gradient on one side of the stationary value scored no marks for the test.

(c) The concept of “rate of change” was not understood by many. Approximately equal

numbers of candidates substituted into $t = 1$ into the expression for y , $\frac{dy}{dt}$ or $\frac{d^2y}{dt^2}$ and so only about one third of the candidates were able to score any marks on this part. Those

who used $\frac{dy}{dt}$ often made careless arithmetic errors when adding three numbers.

Part (d) As in part (c), candidates did not realise which expression to use and perhaps the majority wrongly selected the second derivative. It is a general weakness that candidates do not realise that the sign of the **first** derivative indicates whether a function is increasing or decreasing at a

Q3.

Part (a)(i) was well answered by many candidates, who obtained full marks. Many often went on to obtain $e^x(x^2 + 2x - 3)$ which helped them later in the question. Part (a)(ii) was also well answered but not quite as well as part (a)(i). Many candidates obtained partial credit for reasonable attempts at the product rule but often there was a term missing.

In part (b)(i), the removal of e^x proved to be a major challenge: a large number of candidates chose to take logs (incorrectly) and, although some were lucky in the way they had written out the next step and recovered marks, many generated unmanageable expressions. In part (b)(ii), many correct responses were seen.

Q4.

The evaluations in part (a) were done correctly by most candidates, although some made numerical errors. In part (a)(ii), the request for three significant figures was largely adhered to, although some interpreted this as three decimal places.

Candidates started part (b) well but many were confused as to how to handle the negative signs.

Many attempts to take logs of negative numbers were seen. Many got to $14 \ln\left(\frac{5}{12}\right)$ or $-14 \ln\left(\frac{5}{12}\right)$ and gave this as their answer. Despite this, the vast majority gave the correct answer of 12 to part (b)(ii), some just ignoring their negative sign or saying it must be positive. They apparently did not respond to a negative answer by reviewing their answer to part (b)(i). Very few candidates explained what they were doing with the negative index to obtain the expected answer. In part (c)(i), very few candidates derived the differential equation with confidence. Some made a start by attempting to differentiate the expression $\frac{1}{14}$ for x , although, of those, many made the error of bring down the t together with $\frac{1}{14}$. Others attempted to find an expression for t and differentiate that. There was usually little progress beyond these starting points, although several candidates attempted to convince the examiner that they had found the requested result, with much abuse of the $\ln$ function seen in such attempts.

Some tried to integrate the given differential equation to recover the expression in x , and some very neat solutions were seen this way, although those who did not consider limits or a constant could make no further progress. Again, handling negative signs with logarithmic expressions caused difficulties. Most candidates scored the final mark for

applying the result from part (c)(i), although some did say that $\frac{1}{14}(7) = 2$.

Q5.

- (a) (i) The differentiation was generally well done, though some candidates found the fractional coefficient problematic. Those who wrote $6/3t^5$ were not penalised in this part but writing $6 \cdot \frac{1}{3}$ generally led to errors later in the question. Some tried to avoid the fraction by considering $3V$ throughout the question, making errors, and suffered a heavy penalty.
- (ii) Again, most applied the method correctly and here simplification of coefficients was necessary. A few failed to differentiate $6t$ or omitted it altogether.
- (b) This part was poorly done with many not recognising that the rate of change was $\frac{dV}{dt}$ and substituted $t = 2$ into the expression for V or $\frac{d^2V}{dt^2}$. Quite a lot of candidates made arithmetic errors. A few found two values of the expression and averaged them.
- (c) (i) Again, many failed to evaluate $\frac{dV}{dt}$ in order to verify that a stationary point occurred, but those who did generally obtained a value of zero. It was essential to include a relevant statement to earn both marks.
- (ii) This required evaluation of the second derivative at $t = 1$ or an appropriate test. Candidates who tried to test the gradient on either side of 1 almost invariably failed, as the values used were too far away from the stationary point. A surprising number evaluated $10 - 24 + 6$ to be -20 thus losing the accuracy mark. Some appeared to be guessing and drew wrong conclusions about maxima or minima after evaluating the second derivative.

Q6.

This question also proved to be a good source of marks for many candidates. The examiners expected to see some intermediate evaluation before the printed answer was quoted in part (a)(i) after the substitution of 4 for x in the given equation. Most candidates gave the correct value for the power k although the wrong values $1/2$ and $1/2$ were both seen in part (a)(ii). Candidates were generally able to differentiate the given expression to find the second derivative in part(a)(iii) but some believed that the sign of the x value rather than the sign of the second derivative determined the nature of the stationary point in (iv). In part (b) a significant number of candidates gave the equation of the tangent instead of the normal or used the gradient of the normal as -12 . Finding the integral in part (c)(i) was well answered although, somewhat surprisingly, the integration of -7 caused as many problems as the integration of the other two terms. The final part of the question proved to be beyond many average candidates who forgot to insert the constant of integration in answering the previous part. The hint 'Hence', which should have suggested the use of the answer to (c)(ii) was not always picked up. It was not uncommon to see the equation of the curve given as a linear equation, normally the tangent at P .

Q7.

This question proved to be a good source of marks for many of the candidates. Its structure enabled candidates to progress even if they were unsuccessful with some of the parts.

Part (a)(i) Reasonably well done. Errors were with the derivative of e^{2x} . Common errors were e^{2x} and $2x e^{2x}$. A number of candidates added a '+ c' when they were differentiating.

Part(a)(ii). Usually correct, with errors, if seen, similar to those in the previous part.

Part (b)(i) The majority of candidates answered this correctly.

Part (b)(ii) Very well answered by the majority of candidates.

Part (b)(iii) Again very well answered with many totally correct responses. The major error apart from those who used the incorrect equation $e^{2x} - 5 e^{2x} + 6$ was the evaluation of e^{21n3} which was often seen as 6 rather than 9. Possibly the evaluation of e^{21n2} as 4 was fortuitous for some candidates.

Part (iv) Similar to part (iii) with similar errors, the major one being the evaluation of $4e^{21n3}$ as $4 \times 6 = 24$. This resulted in an answer of -6 and hence loss of accuracy marks.

Q8.

(a) Once candidates realised that the lengths in centimetres of the sides of the box were $24 - 2x$, $9 - 2x$ and x , they were usually able to multiply out to give the printed answer for the volume. A large number of weaker candidates found this part of the question very difficult and made little or no effort to obtain the printed answer. (a)(i) The expression for dV/dx was usually correct.

(ii) This part was answered too casually by many candidates who simply divided their previous answer by 12 and wrote " $= 0$ " to comply with the printed answer. It was necessary to see that the candidate realised that it is

$$\frac{dV}{dx} = 0$$
 necessary for $\frac{dV}{dx}$ for stationary points to occur, before dividing their expression for dV/dx by 12.

(iii) Most candidates scored the marks for solving the quadratic equation, although a few who chose to use the formula, rather than factorising, made slips in their arithmetic.

(c) It was pleasing to see that most candidates were able to find a second derivative. However, this was often given as $2x - 11$, since many had previously divided dV/dx by 12 and were using this modified expression. Some candidates did not appear to understand the second derivative test for maxima and minima; they

$$\frac{d^2V}{dx^2} = 24x - 132 = 0$$
 wrote things such as " $\frac{d^2V}{dx^2}$ when $x = 5.5$ and since this is positive the stationary point is a minimum".

Q9.

This question was answered very well by many candidates, and evidently the basic processes of differentiation and integration had been well rehearsed.

Part (a)(i) Most candidates substituted $x = 3$ into the given cubic equation and found $p(x)$ to be zero. Some, however, lost a mark as they did not indicate that $p(3) = 0$ implied that x

– 3 was a factor.

Part (a)(ii) The coordinates (3, 0) were often seen but some forgot to give the y coordinate

Part (b)(i) Almost all candidates differentiated correctly, which was pleasing.

Part (b)(ii) Most candidates realised the need to equate dy/dx to zero. However not all were able to solve the resulting equation. Some of those who did solve the quadratic offered both solutions, 1 **and** $7/3$. Many found the second derivative in order to ascertain which point was the minimum point M . This was unnecessary as the coordinates of A (1, 0) were clear from the diagram.

Part (c) The second derivative was almost always found correctly followed by the substitution of $x = 1$.

Part (d)(i) The integration was carried out very well. Occasionally $\frac{3x^4}{4}$ was seen but this was rare.

Part (d)(ii) Most candidates used the correct limits of 0 and 1. However many arithmetic errors occurred.

When trying to combine the fractions so that the correct value of the integral $-\frac{11}{12}$ was not seen very often. Of those who did integrate correctly, many still lost the last mark for not stating that the area of the shaded region was $11/12$ and the negative value of the integral arose because the region was below the x -axis. It was not acceptable to simply

write Area = $-\frac{11}{12} = \frac{11}{12}$.

Q10.

As in the previous question, many candidates arrived at the correct answer after a considerable algebraic struggle. Errors often occurred in the course of the manipulation, but some candidates who coped successfully with the cubing of $2 + h$ then forgot to add a further $2 + h$ and to subtract 10, though in many cases they had previously declared their intention of doing these things. Unfortunately a high proportion of candidates who reached an answer of the required form in part (a) did not use it properly in part (b), resorting instead to a direct differentiation of the given function f or of their answer to part (a).